

AI-Driven Predictive Ontology for Cold Atmospheric Plasma Therapy in Immune Diseases: A Multi-Stage Framework from RONS Generation to Treatment Mechanisms

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(Names with versions only, sorted by contribution)*

Abstract

Cold Atmospheric Plasma (CAP) therapy has emerged as a promising treatment modality for immune-mediated diseases, yet the complex causal relationships between plasma-generated Reactive Oxygen and Nitrogen Species (RONS) and therapeutic outcomes remain poorly understood. We present a novel AI-driven predictive ontology framework that models the complete causal chain from plasma device parameters to disease treatment mechanisms through five interconnected stages: (1) device conditions to RONS generation, (2) RONS exposure to tissue penetration, (3) penetration to cellular responses, (4) cellular changes to signaling pathway activation, and (5) signaling to therapeutic mechanisms. Our framework integrates OWL-based semantic reasoning with machine learning models, leveraging data from public databases including KEGG, Reactome, STRING, and CTD. We demonstrate the framework's utility through a case study on atopic dermatitis, showing how AI-assisted hypothesis generation and ontology reasoning can predict treatment mechanisms. This work establishes a foundation for systematic plasma medicine research and personalized treatment optimization.

1 Introduction

Cold Atmospheric Plasma (CAP) represents a revolutionary approach to treating various medical conditions, particularly immune-mediated diseases such as atopic dermatitis, psoriasis, rheumatoid arthritis, and asthma [1, 2]. CAP generates a complex mixture of Reactive Oxygen and Nitrogen Species (RONS), including hydroxyl radicals (OH), hydrogen peroxide (H₂O₂), superoxide (O₂⁻), nitric oxide (NO), and peroxynitrite (ONOO⁻), which interact with biological tissues to produce therapeutic effects [3].

Despite growing clinical evidence supporting CAP efficacy, the field faces significant challenges in understanding and predicting treatment outcomes. The relationship between plasma device parameters and clinical efficacy involves multiple interconnected biological processes spanning different scales—from molecular interactions to tissue-level responses.

Current research often examines isolated segments of this causal chain, limiting our ability to optimize treatment protocols systematically.

The emergence of Large Language Models (LLMs) and AI-driven scientific discovery tools presents unprecedented opportunities to address these challenges. AI systems can assist in hypothesis generation, literature synthesis, and complex reasoning across interdisciplinary domains [4]. Combined with formal ontology frameworks, AI can enable systematic knowledge representation and inference about plasma-biology interactions.

In this paper, we present a comprehensive AI-driven predictive ontology framework for plasma medicine that models the complete therapeutic pathway through five stages: (1) plasma device parameters → RONS generation, (2) RONS → tissue penetration, (3) penetration → cellular responses, (4) cellular changes → signaling pathways, and (5) signaling → treatment mechanisms. Our key contributions include:

- A formal OWL ontology capturing the domain knowledge of plasma medicine, including RONS species, penetration mechanisms, cellular responses, and disease-specific treatment pathways
- A five-stage prediction pipeline integrating physics-based models, machine learning, and ontology reasoning
- Integration with major biological databases (KEGG, Reactome, STRING, CTD) for evidence-based predictions.
- A demonstration of AI-assisted research methodology through a case study on atopic dermatitis treatment.

2 Related work

2.1 Plasma Medicine Research

Cold atmospheric plasma has been studied extensively for wound healing, cancer treatment, and dermatological applications [5]. Bekeschus et al. demonstrated that plasma-derived RONS modulate immune cell behavior through redox signaling [6]. However, most studies focus on specific aspects of the plasma-biology interaction without providing integrated predictive frameworks.

2.2 Biomedical Ontologies

Formal ontologies have proven valuable for knowledge representation in biomedicine. Gene Ontology (GO), Disease Ontology (DO), and pathway databases like KEGG and Reactome provide structured vocabularies for biological processes [7, 8]. The Chemical Entities of Biological Interest (ChEBI) ontology includes some reactive species, but no comprehensive ontology exists for plasma medicine applications.

2.3 AI in Scientific Discovery

Recent advances in LLMs have enabled AI-assisted scientific research, from literature review to hypothesis generation [9]. The AI Scientist framework and similar systems demonstrate the potential for automated research workflows [10]. Our work extends these approaches to the specific domain of plasma medicine, combining LLM capabilities with formal ontological reasoning.

3 Methodology

Our framework consists of three main components: (1) a domain ontology for plasma medicine, (2) a five-stage prediction pipeline, and (3) AI-assisted reasoning and hypothesis generation.

3.1 Ontology Design

We developed the Plasma Medicine Ontology (PMO) using OWL 2 DL, following established ontology engineering practices. The ontology comprises seven main modules: Core Classes: Our class hierarchy includes PlasmaDevice (DBD, PlasmaJet), RONS with subclasses ROS (OH, H₂O₂, O₂⁻, O₃, ¹O₂) and RNS (NO, NO₂, ONOO⁻, HNO₂), PenetrationMechanism, CellularResponse, SignalingPathway, Immune Disease, and Treatment Mechanism.

Object Properties: Key relationships include generates (PlasmaDevice → RONS), penetratesVia (RONS → PenetrationMechanism), induces (RONS → CellularResponse), activates (CellularResponse → Signaling Pathway), and treats (Treatment Mechanism → Immune Disease).

Data Properties: We capture quantitative attributes including hasLifetime (duration in ns-ms), hasPenetrationDepth (μm), hasConcentration (μM or ppm), hasDiffusionCoefficient (cm²/s), and hasFoldChange for gene expression.

3.2 Five-Stage Prediction Pipeline

Stage 1 - Device to RONS Prediction: Given plasma device parameters (type, voltage, frequency, gas composition, flow rate, treatment distance), we predict RONS concentrations using a combination of physics-based plasma chemistry models and machine learning regressors trained on published experimental data. The model outputs concentrations for each RONS species with confidence intervals.

Stage 2 - RONS to Penetration: We model RONS transport into tissue using reaction-diffusion equations:

$$\partial C / \partial t = D \nabla^2 C - kC + S$$

where C is concentration, D is diffusion coefficient, k is reaction rate, and S is source term. Different penetration mechanisms (direct diffusion, lipid peroxidation chain, gap junction communication, calcium waves, paracrine signaling) are modeled with mechanism-specific parameters.

Stage 3 - Penetration to Cellular Response: Using the ontology mappings and database integration, we predict cellular responses based on penetrated RONS concentrations. The prediction involves:

- Gene expression changes via RONS-gene relationship mappings from CTD
- Protein modifications through redox-sensitive residue analysis
- Cellular function changes (viability, proliferation, apoptosis) via dose-response models

Stage 4 - Cellular Response to Signaling: Pathway enrichment analysis identifies activated signaling cascades. We integrate:

- KEGG pathway mappings for metabolic and signaling pathways
- Reactome for detailed reaction networks
- STRING for protein-protein interactions

Key pathways include NF- κ B (inflammation), Nrf2 (antioxidant response), MAPK (stress response), and JAK-STAT (cytokine signaling).

Stage 5 - Signaling to Treatment Mechanism: The final stage maps activated pathways to disease-specific treatment mechanisms using:

- Disease-pathway associations from DisGeNET
- Ontology reasoning for mechanism inference
- Literature evidence scoring from PubMed

3.3 AI-Assisted Reasoning

We employ LLMs (Claude, GPT-4) for several tasks:

- Literature synthesis and hypothesis generation
- SPARQL query formulation for ontology queries
- Natural language interface for the prediction system
- Explanation generation for prediction results

The AI interaction follows a structured protocol ensuring reproducibility, with all prompts and responses logged for transparency.

4 Data sources and integration

Our framework integrates multiple public databases to support evidence-based predictions:

4.1 Biological Databases

KEGG (Kyoto Encyclopedia of Genes and Genomes): We utilize KEGG pathway maps for signaling pathway analysis, particularly hsa04630 (JAK-STAT), hsa04010 (MAPK), hsa04064 (NF- κ B), and disease-specific pathways for our target conditions.

Reactome: Provides detailed biochemical reaction networks, including redox reactions involving RONS and cellular antioxidant systems.

STRING: Protein-protein interaction data enables network-based analysis of cellular response propagation.

CTD (Comparative Toxicogenomics Database): Chemical-gene-disease relationships provide evidence for RONS effects on gene expression and disease associations.

Gene Ontology: Functional annotations support cellular response categorization.

4.2 Literature Data

We systematically collected plasma medicine literature from PubMed using queries combining "cold atmospheric plasma" with target disease terms. AI-assisted extraction identified:

- Device parameters and corresponding RONS measurements
- Cellular response data from in vitro studies
- Clinical outcome data from human studies

4.3 Data Processing Pipeline

Raw data from databases undergoes standardization:

- Entity mapping to ontology classes via exact and fuzzy matching
- Unit normalization for quantitative values
- Confidence scoring based on evidence type and source reliability

Database	Data Type	Records Used	License
KEGG	Pathways	~350 pathways	Academic
Reactome	Reactions	~2,500 reactions	CC BY 4.0
STRING	PPIs	~20,000 interactions	CC BY 4.0
CTD	Chemical-Gene	~15,000 relations	Public
PubMed	Literature	~500 papers	Public

Table 1: Summary of integrated data sources

5 Case study: atopic dermatitis

We demonstrate our framework through a case study on atopic dermatitis (AD), a chronic inflammatory skin disease affecting up to 20% of children and 3% of adults worldwide [11].

5.1 Prediction Scenario

Input Parameters:

- Device: DBD plasma
- Voltage: 10 kV, Frequency: 10 kHz
- Gas: Ambient air, Treatment distance: 2 mm
- Treatment time: 60 seconds

5.2 Stage-by-Stage Results

Stage 1 Output: Predicted RONS concentrations:

- H₂O₂: 50-100 μ M (confidence: 0.85)
- NO: 10-30 μ M (confidence: 0.78)
- OH: transient, $\sim$ 1 μ M peak (confidence: 0.72)
- O₃: 5-15 ppm in gas phase (confidence: 0.80)

Stage 2 Output: Penetration profile analysis indicates:

- H₂O₂ penetrates to $\sim$ 200 μ m via aquaporins and direct diffusion
- NO penetrates deeper ($\sim$ 500 μ m) due to lipophilicity
- Secondary signaling (lipid peroxidation products, calcium waves) extends effects to $\sim$ 1 mm

Stage 3 Output: Predicted cellular responses:

- Keratinocyte: Nrf2 activation (2.5x), mild oxidative stress response
- Fibroblast: Increased collagen synthesis genes
- Immune cells: Th2 cytokine reduction predicted

Stage 4 Output: Pathway activation:

- Nrf2-ARE pathway: Strong activation ($p < 0.001$)
- NF- κ B pathway: Moderate suppression

- JAK-STAT pathway: IL-4/IL-13 signaling reduced
- TGF- β pathway: Moderate activation (barrier repair)

Stage 5 Output: Treatment mechanisms for AD:

- Anti-inflammatory: NF- κ B suppression reduces pro-inflammatory cytokines
- Barrier repair: Filaggrin expression increase predicted
- Immunomodulation: Th2 $\rightarrow$ Th1 shift suggested
- Antimicrobial: RONS direct effects on *S. aureus*

5.3 Validation Against Literature

Our predictions align with published clinical observations:

- Isbary et al. [12] reported improved AD symptoms with similar DBD parameters
- Molecular studies confirm Nrf2 activation by plasma treatment [13]
- Barrier function improvement observed in clinical trials [14]

The framework successfully identifies the key therapeutic pathways while providing mechanistic explanations grounded in ontological knowledge. (Deployed Website: <https://plasma-medicine-api.fly.dev/>)

6 Discussion and limitations

6.1 Key Findings

Our AI-driven ontology framework demonstrates several advantages:

- Systematic integration of multi-scale biological knowledge
- Explainable predictions through ontological reasoning
- AI-assisted hypothesis generation accelerates research workflows

The case study shows the framework can generate clinically relevant predictions that align with published evidence, suggesting utility for treatment optimization.

6.2 Limitations

Several limitations should be acknowledged:

Data Availability: Quantitative RONS measurement data under standardized conditions remains limited. Model predictions carry uncertainty that propagates through the pipeline.

Biological Complexity: Our five-stage model necessarily simplifies complex biological interactions. Feedback loops, individual variability, and microenvironment effects are not fully captured.

Validation: While predictions align with literature, prospective experimental validation is needed. The framework generates hypotheses requiring wet-lab confirmation.

AI Reliability: LLM-generated content requires careful verification. We implemented human-in-the-loop validation for all AI-assisted components.

6.3 Broader Impact

Positive impacts include accelerating plasma medicine research and potentially enabling personalized treatment protocols. Negative impacts could arise from over-reliance on predictions without experimental validation, potentially leading to suboptimal treatments. We emphasize that this framework supports, rather than replaces, rigorous experimental science.

7 Conclusion

We presented an AI-driven predictive ontology framework for plasma medicine that models the complete causal chain from device parameters to treatment mechanisms. By integrating formal ontological reasoning with machine learning and leveraging public biological databases, our framework enables systematic prediction and explanation of plasma therapy effects.

The atopic dermatitis case study demonstrates the framework's ability to generate clinically relevant, mechanistically grounded predictions. This work establishes a foundation for AI-assisted plasma medicine research, with potential applications in treatment protocol optimization and personalized medicine.

Future work will focus on expanding the ontology coverage, incorporating patient-specific factors, and validating predictions through prospective experimental studies. We will release the ontology and prediction tools to support the plasma medicine research community.

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AI Co-Scientist Challenge Korea Paper Checklist

1. Claims

Answer: [Yes]

Justification: The abstract and introduction clearly state our contributions: a predictive ontology framework for plasma medicine with five-stage pipeline.

2. Limitations

Answer: [Yes]

Justification: Section 6.2 discusses data availability, biological complexity simplification, validation needs, and AI reliability limitations.

3. Theory Assumptions and Proofs

Answer: [N/A]

Justification: This paper presents a framework and methodology rather than theoretical proofs.

4. Experimental Result Reproducibility

Answer: [Yes]

Justification: We describe the methodology, data sources, and provide the case study parameters sufficient for reproduction.

5. Open access to data and code

Answer: [Yes]

Justification: We will release the ontology files and code upon acceptance. All external databases used are publicly available.

6. Experimental Setting/Details

Answer: [Yes]

Justification: Section 5 provides detailed parameters for the case study including device settings and prediction outputs.

7. Experiment Statistical Significance

Answer: [N/A]

Justification: This is primarily a framework paper with a demonstration case study rather than statistical experiments.

8. Experiments Compute Resources

Answer: [Yes]

Justification: Predictions were generated using standard computing resources (CPU-based, no GPU required for ontology reasoning).

9. Code of Ethics

Answer: [Yes]

Justification: We followed ethical guidelines. No human subjects involved. AI usage is transparently documented.

10. Broader Impacts

Answer: [Yes]

Justification: Section 6.3 discusses both positive impacts (accelerating research) and potential negative impacts (over-reliance on predictions).

11. Safeguards

Answer: [N/A]

Justification: The framework poses no risks for misuse. It is designed for legitimate medical research support.

12. Licenses for existing assets

Answer: [Yes]

Justification: Table 1 lists all database sources with their licenses. All are appropriately licensed for research use.

13. New Assets

Answer: [Yes]

Justification: The ontology will be released under CC BY 4.0 license with documentation.

14. Crowdsourcing and Human Subjects

Answer: [N/A]

Justification: No crowdsourcing or human subjects research involved.

15. IRB Approvals

Answer: [N/A]

Justification: No human subjects research requiring IRB approval.